

Topic: Introduction to Relational Algebra

Q. What is Relational Algebra in DBMS? Explain its basic concepts and categorize the primary relational algebra operators.

 **Definition to Remember**

Relational Algebra is a **procedural query language** that forms the theoretical foundation of SQL. It specifies both *what* data is needed and *how* to retrieve it through a sequence of operations on relations (tables), where every operation takes one or two relations as input and produces a new relation as output.

Introduction to Relational Algebra

Concept	Explanation
Input/Output	Every operation takes 1 or 2 relations (tables) as input and produces a new relation as output.
Closure Property	Since output is always a relation, operations can be nested and chained in sequence.
No Side Effects	Operations only query data — they never modify, insert, or delete from the original tables.

2. Relational Algebra Operators

A. Basic (Fundamental) Operators

(All other operators can be derived from these)

Operator	Symbol	Description
Selection	σ	Extracts specific rows satisfying a condition
Projection	π	Extracts specific columns , discarding others
Union	$\cup$	Combines rows from two relations, removes duplicates
Set Difference	$-$	Rows in first relation NOT in the second
Cartesian Product	$\times$	Combines every row of A with every row of B
Rename	ρ	Renames a relation or its attributes

B. Derived Operators

(Combinations of basic operators for convenience)

Operator	Symbol	Description
Intersection	$\cap$	Rows present in both relations (derived from -)
Join	$\bowtie$	Combines related tuples from two relations (Cartesian $\times$ Selection)
Division	$\div$	Answers "for all" queries (e.g., students who took all courses)

★ Must-Write Points (for 10 marks)

1. Relational Algebra is a procedural query language — the mathematical foundation of SQL.
2. Every operation takes 1 or 2 relations as input and always outputs a new relation (**Closure Property**).
3. Operations do not modify the original tables — they are read-only.
4. Basic operators: Selection (σ), Projection (π), Union ($\cup$), Set Difference ($-$), Cartesian Product ($\times$), Rename (ρ).
5. Derived operators: Intersection ($\cap$), Join ($\bowtie$), Division ($\div$).
6. Derived operators can be expressed using combinations of basic operators.
7. Understanding Relational Algebra helps understand how the DBMS engine executes and optimizes SQL queries.

⚡ Quick Recall

Procedural Language $\rightarrow$ Closure Property $\rightarrow$ Basic ($\sigma, \pi, \cup, -, \times, \rho$) $\rightarrow$ Derived ($\cap, \bowtie, \div$) $\rightarrow$ Foundation of SQL